LOAN DEFAULT PREDICTION

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Problem & Motivation

Why predict default?

1

The business problem

Undetected loan defaults can translate into significant financial loss for lending institutions.

2

Why rules are limited

Traditional scoring may miss non-linear relationships among income, debt, credit history and repayment behaviour.

3

The ML objective

Flag high-risk applicants before disbursement while keeping the reasoning interpretable for credit analysts.

Core challenge

- Default is the minority class, so accuracy alone can be misleading.
- The model should catch genuine defaulters without creating excessive false alarms.
- Predictions should support — not replace — human lending decisions.

Objectives

Project goals

01

Prepare data

Clean, encode and scale real-world financial data.

02

Engineer features

Create risk-relevant variables such as debt-to-income ratio and credit-score bands.

03

Compare models

Benchmark Logistic Regression against Random Forest.

04

Evaluate correctly

Use precision, recall, F1-score and ROC-AUC alongside accuracy.

05

Explain risk

Identify influential borrower attributes for analyst interpretation.

06

Support decisions

Demonstrate how predictive analytics can strengthen lending risk management.

Dataset & Key Features

5,000 loan applications

5,000
Loan applications

~29%
Default rate

80/20
Train / test split

10
Model features

Applicant / loan variables

- Age and annual income
- Loan amount
- Credit score
- Employment years
- Debt-to-income ratio
- Number of credit lines
- Home ownership and loan purpose
- Prior default history

Target

`loan_default`

Binary classification target: default vs. no default.

Preprocessing

Categorical: label-encoded for home ownership and purpose

Numeric: StandardScaler applied for Logistic Regression

Why this dataset

A ~29% default rate reflects a realistic, imbalanced lending population — close enough to real portfolios that accuracy alone cannot be trusted as a metric.

This is what motivates evaluating models on precision, recall and F1-score, not accuracy alone.

Methodology

Data to decision

01	02	03	04	05	06
Data Collection	**Pre-processing**	**EDA**	**Model Building**	**Evaluation**	**Interpre-tation**
Borrower + loan attributes	Missing values, encoding, scaling	Correlations & default patterns	LogReg + Random Forest	Accuracy, recall, F1, ROC-AUC	Feature importance, error analysis

Class imbalance was addressed with `class_weight='balanced'` in both classifiers.

A systematic six-stage workflow — from raw applicant data to an interpretable, evaluated model — mirrors the process used in production credit-risk systems.

05 · TWO COMPLEMENTARY CLASSIFIERS

Model Implementation

Logistic Regression + Random Forest

LR

Logistic Regression

- Interpretable baseline
- Trained on standardized features
- class_weight='balanced'
- Predicts default probability
- Higher recall and ROC-AUC in this study

RF

Random Forest

- Captures non-linear interactions
- Trained on unscaled features
- 300 trees, max_depth=8
- class_weight='balanced'
- Higher accuracy and precision in this study

Core Python stack

Pandas • NumPy • Scikit-learn • Matplotlib • Seaborn

Model Results

Held-out 20% test set

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.706	0.496	0.691	0.578	0.792
Random Forest	0.735	0.539	0.619	0.576	0.782

0.691

Best recall • Logistic Regression

0.735

Best accuracy • Random Forest

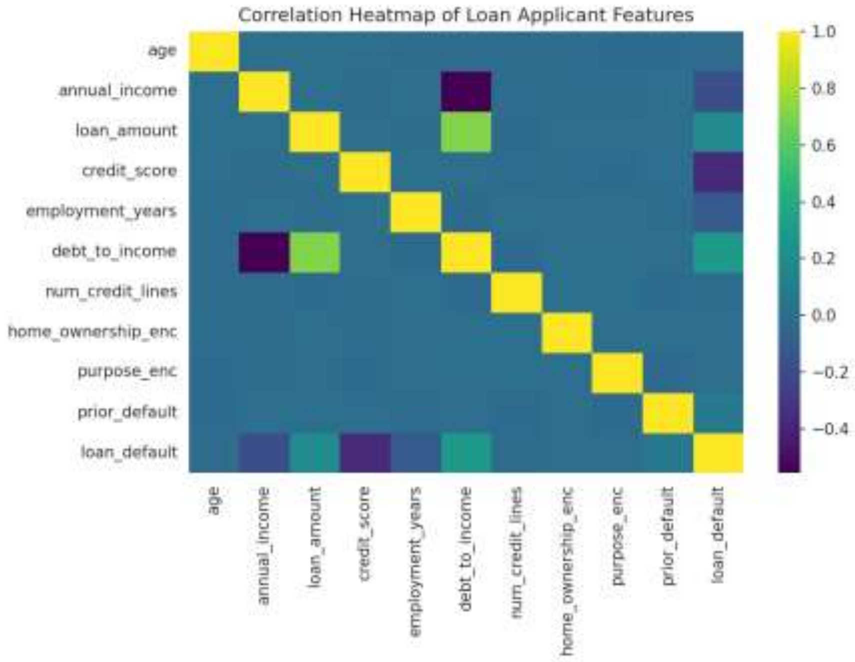
0.792

Best ROC-AUC • Logistic Regression

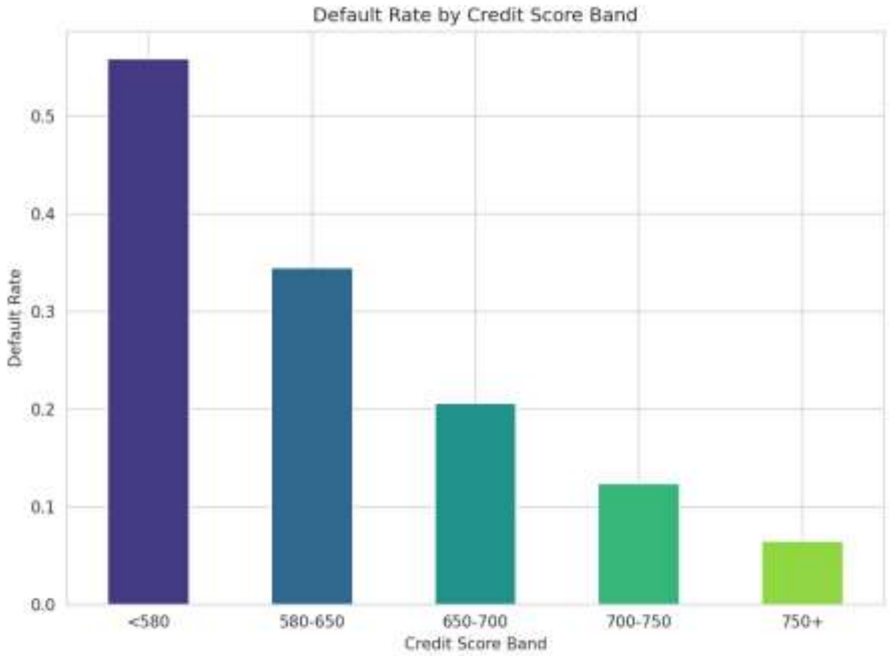
Practical choice: Logistic Regression is favoured when catching true defaulters (recall) is prioritized.

Exploratory Analysis

Correlation & credit risk



Correlation heatmap

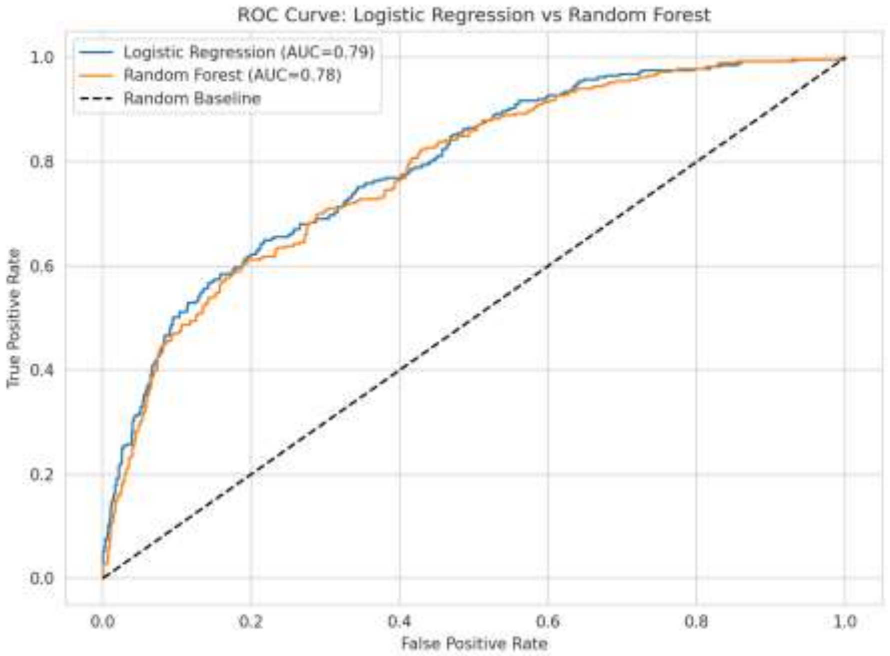


Default rate by credit-score band

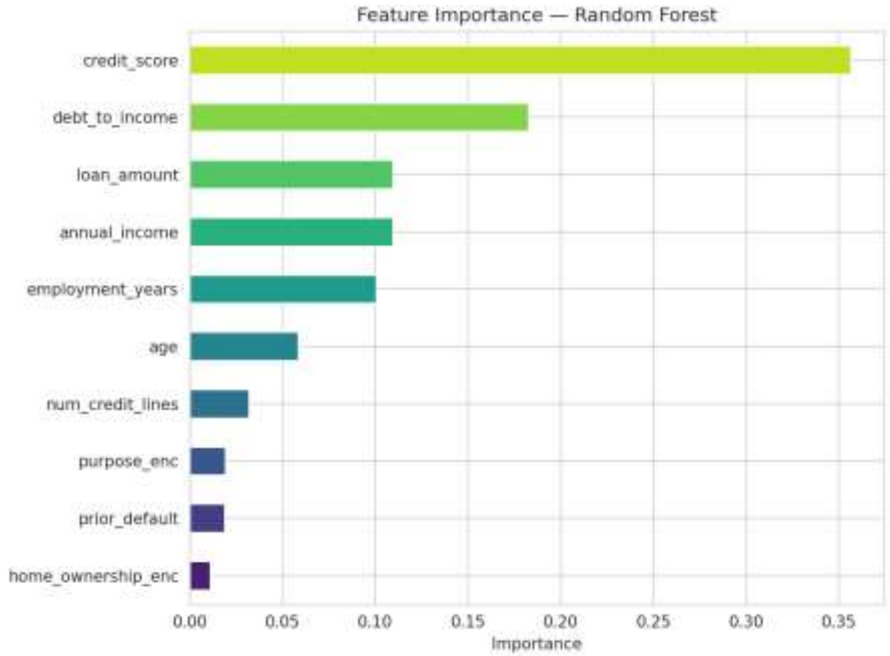
Debt-to-income ratio and prior default history are positively associated with default, while credit score is negatively associated. Default rates fall sharply as credit score improves.

Model Quality & Feature Importance

ROC-AUC and drivers of risk



ROC-AUC ≈ 0.79 for both models

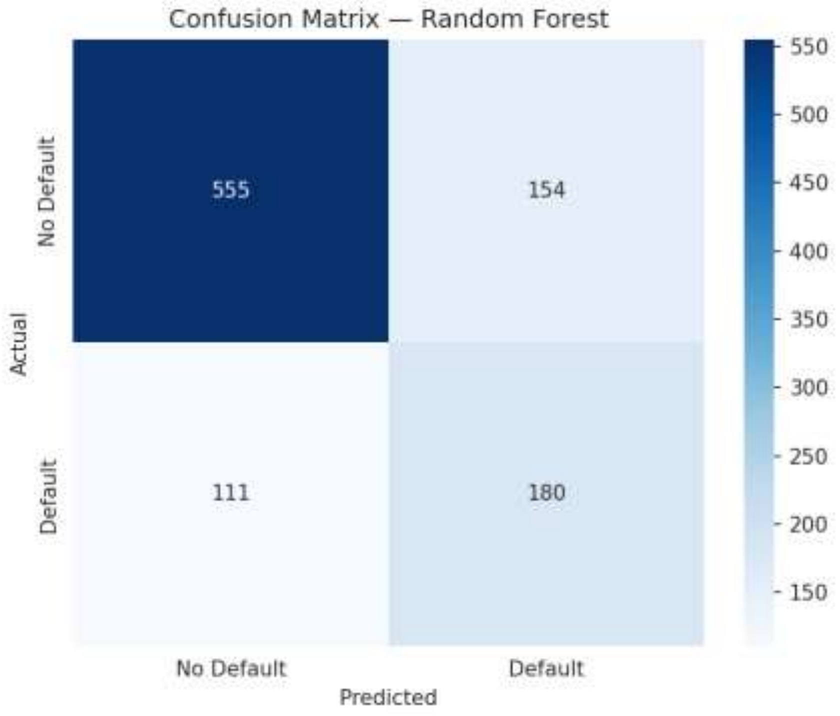


Random Forest: strongest predictors

Both models perform substantially better than the random baseline. Random Forest feature importance highlights credit score, debt-to-income ratio and loan amount among the most influential variables.

Error Analysis

Random Forest confusion matrix



555
Correct non-defaults

180
Correct defaults

154
False default flags

111
Missed defaults

Key limitation

Random Forest correctly identifies most non-defaulters, but its lower recall means a meaningful share of actual defaulters are missed.

Conclusion

Key takeaways

- The project implemented a complete data-science workflow for loan default prediction.
- Both Logistic Regression and Random Forest achieved ROC-AUC close to 0.79.
- Credit score, debt-to-income ratio and prior default history emerged as the strongest risk signals.
- Because defaults are a minority class, recall, precision and F1-score are more informative than accuracy alone.
- The model is intended to support human lending decisions with interpretable evidence.

Overall conclusion

Logistic Regression is the practical choice in this study when the priority is catching more true defaulters.

Future Scope

Where this project goes next

1 Class imbalance

Apply SMOTE or cost-sensitive learning to improve minority-class recall.

2 Advanced ensembles

Benchmark XGBoost / LightGBM and ensemble stacking.

3 Explainable AI

Use SHAP values to explain individual applicant predictions.

4 Deployment

Build a Streamlit or Flask dashboard for real-time risk scoring.

Thank You !